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Department of CSE

B.Tech. CSE Trimester -5

2520030537
Monisha Raghini Chelle

Cloud Infrastructure and Services

25CS2206E

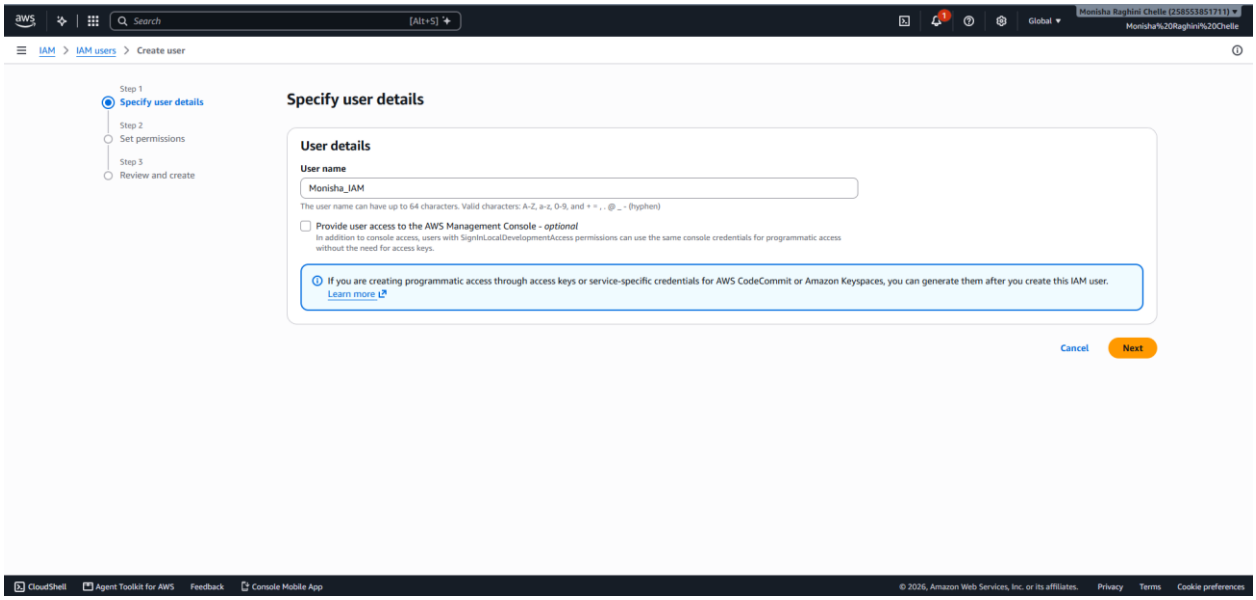
LAB Skill

2026

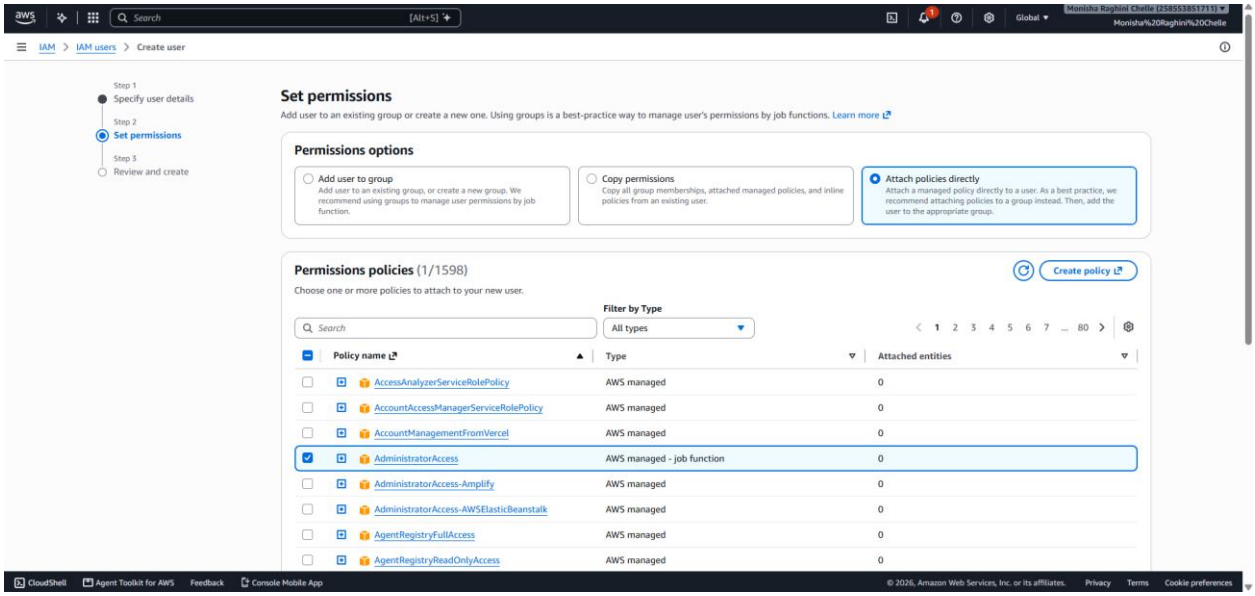
Skill -1

AIM: Create an IAM user with programmatic access only, download the access key and secret key, and document the steps with screenshots. Then set up a billing alert for the account so that an email is received when the Free Tier limit is about to be crossed.

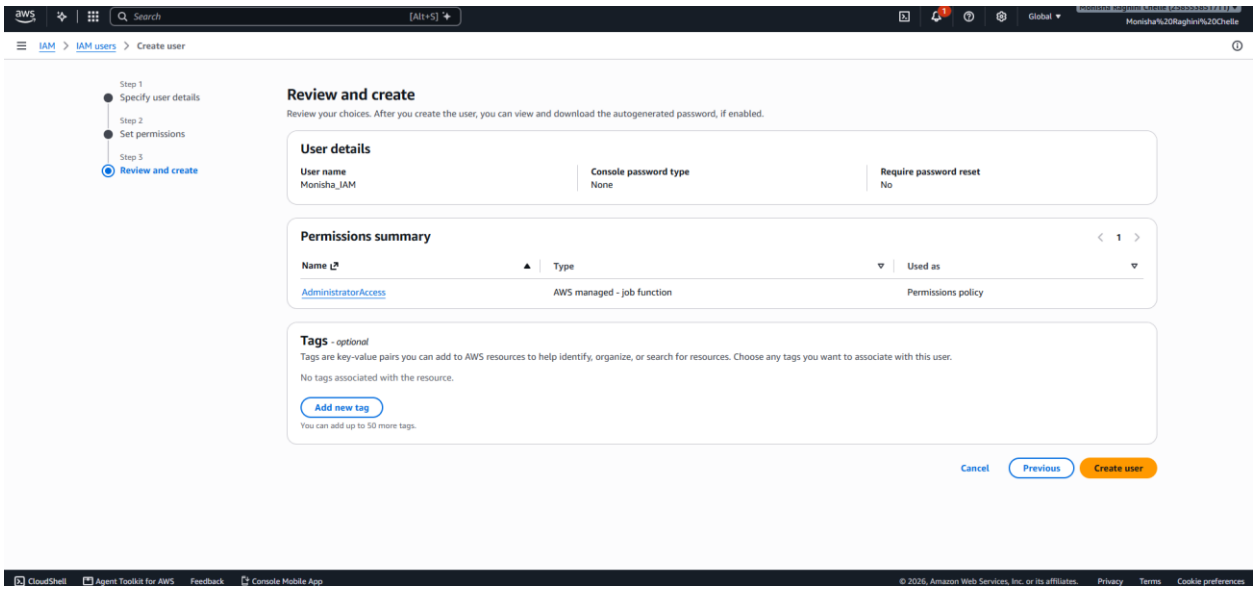
Description:



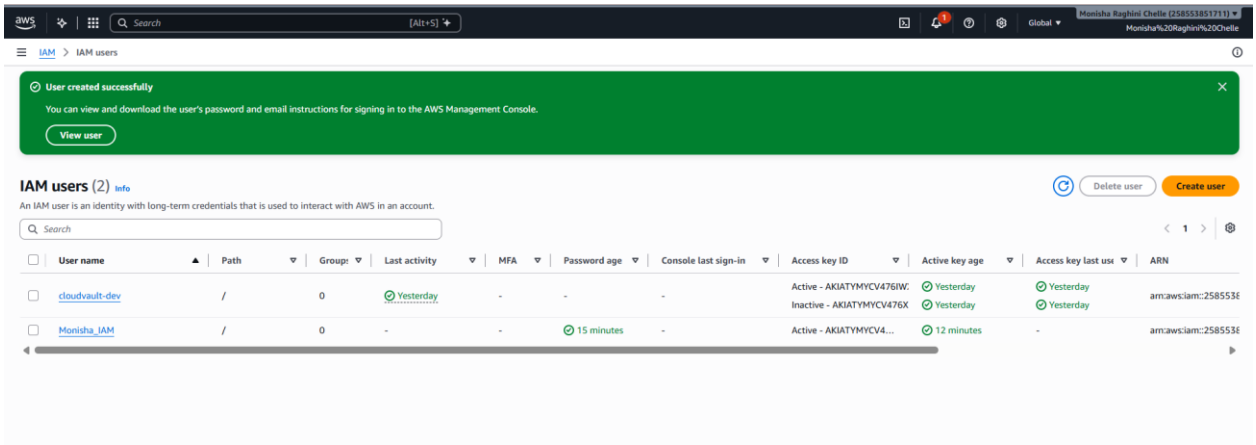
Description: *Entered the username "**Monisha_IAM**" in the User name field and kept the AWS Management Console access option unchecked to create an IAM user without console login access.*



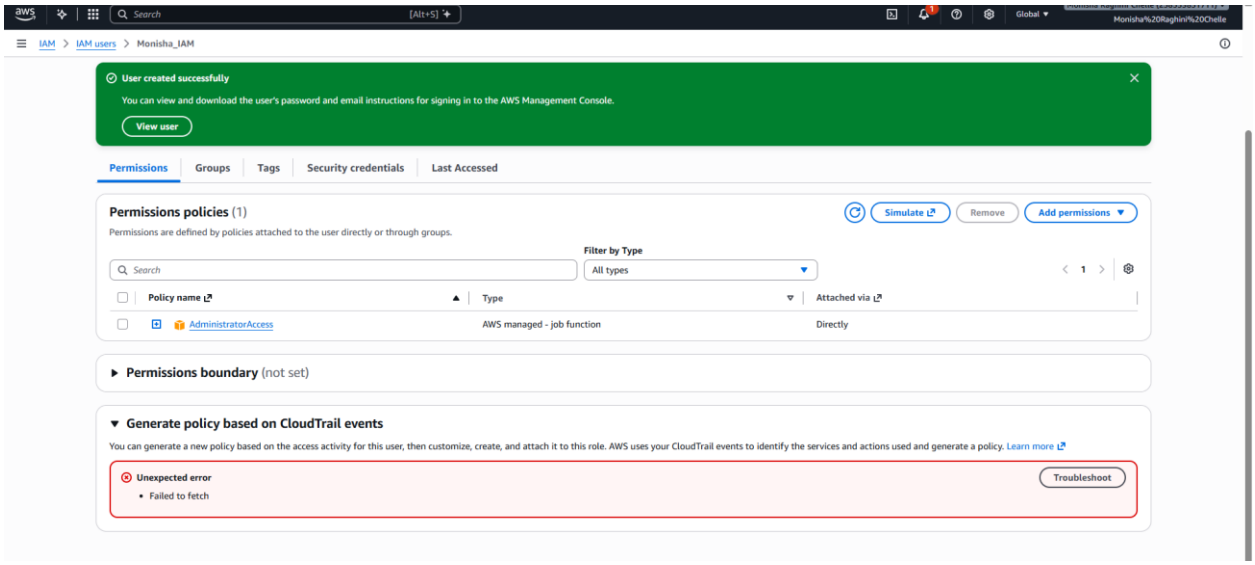
Description: *Selected the* ***Attach policies directly*** *option and chose the* ***AdministratorAccess*** *policy to assign permissions to the IAM user.*



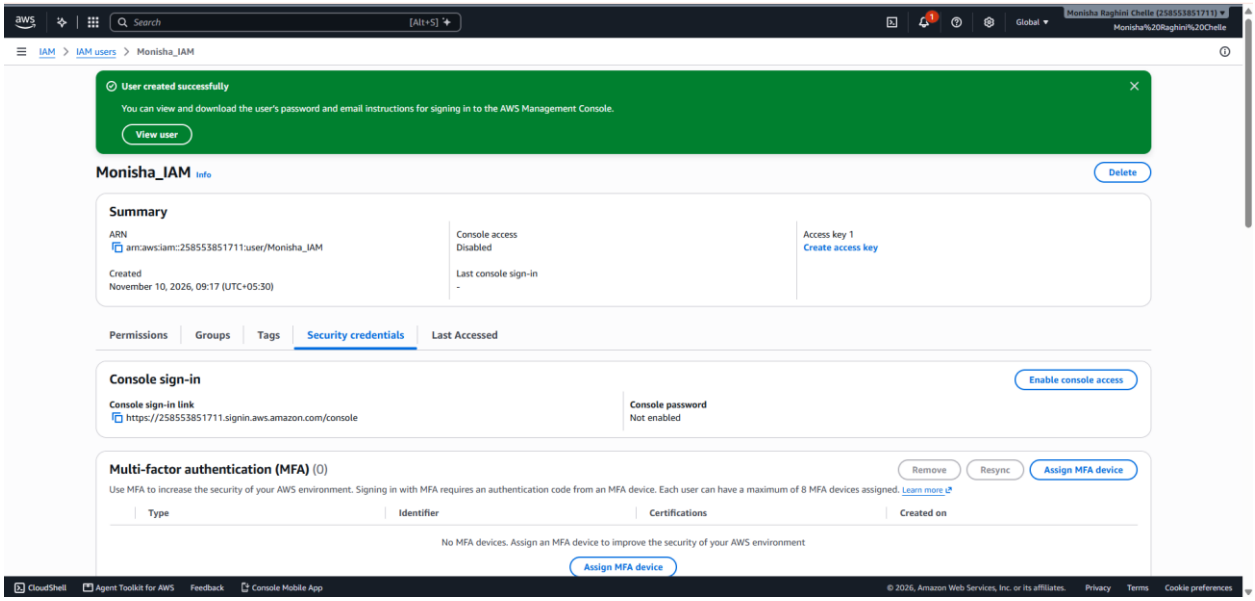
Description: *Reviewed the IAM user details and the assigned AdministratorAccess policy. Confirmed that the console password type is set to None before creating the user.*



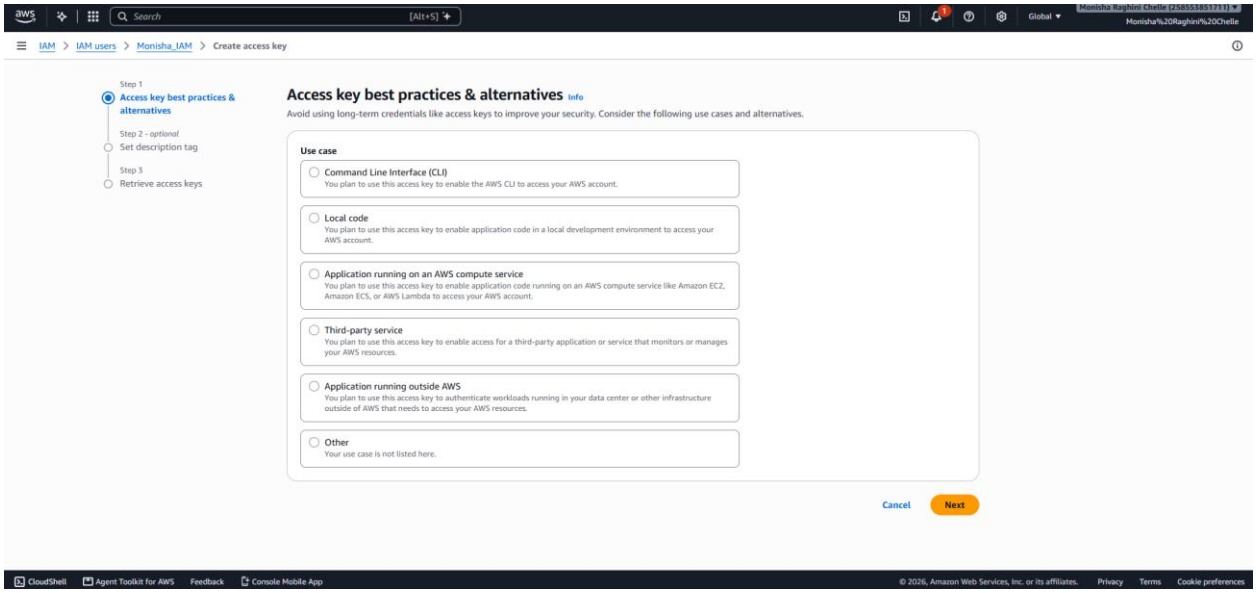
Description: *The IAM user" **Monisha_IAM"** was created successfully and is displayed in the IAM Users list. The user has an active access key.*



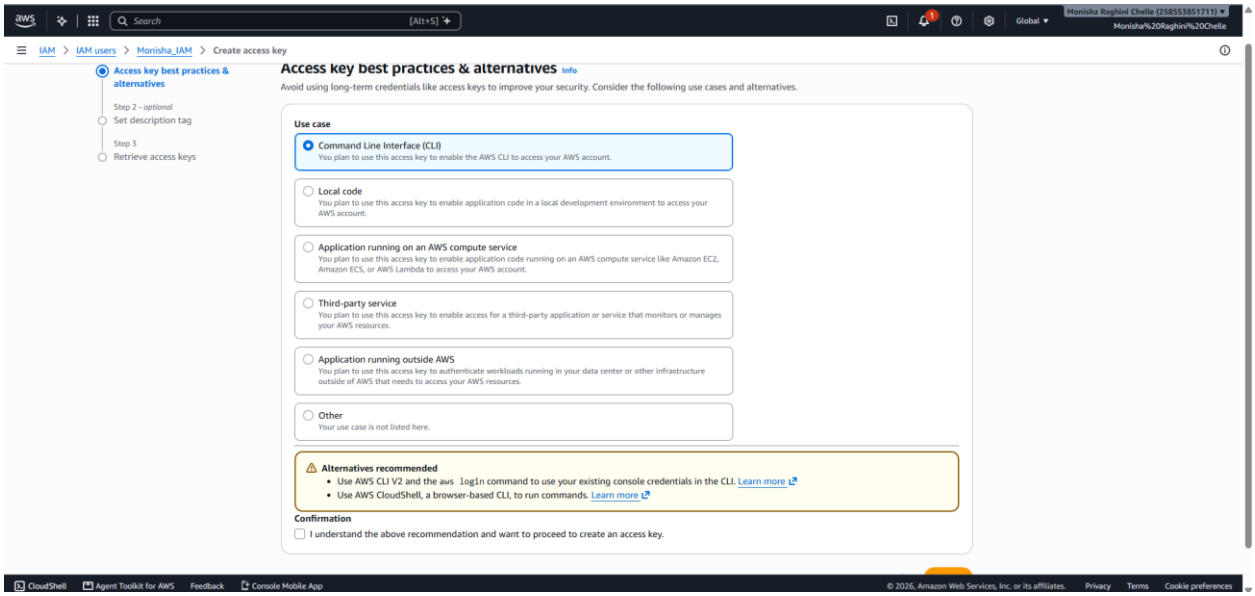
Description: *Opened the IAM user details and checked the Permissions tab. The AdministratorAccess policy is displayed as directly attached to the user.*



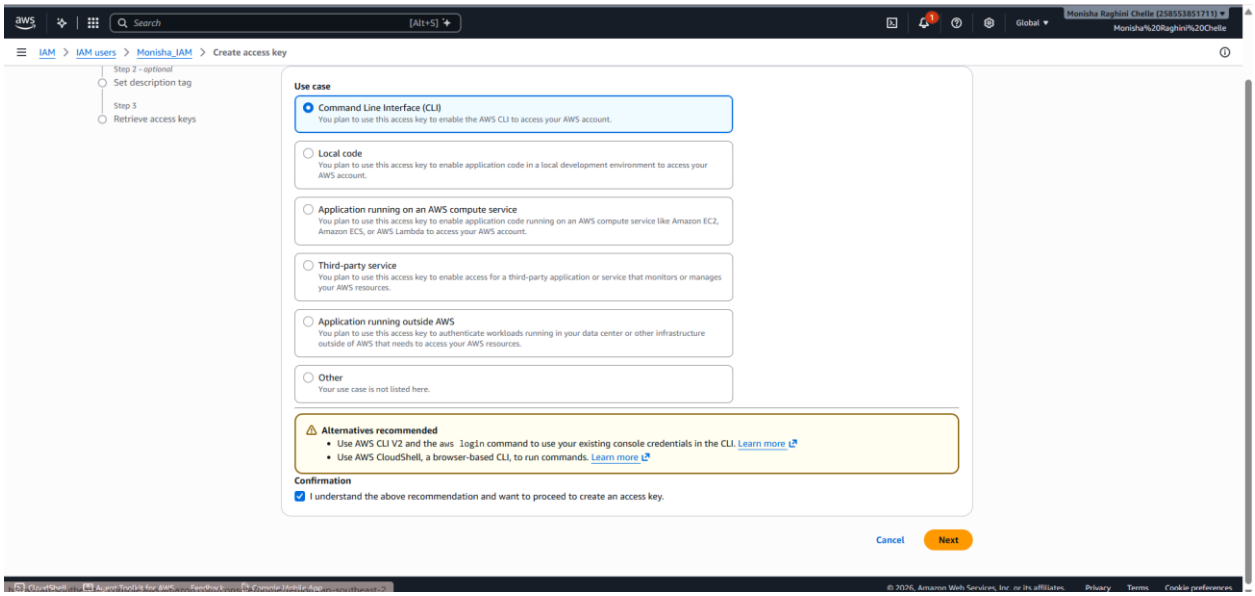
Description: *Opened the Security credentials tab of the" **Monisha_IAM"** user and verified that console access is disabled and the console password is not enabled. The page also displays the access key and MFA settings.*



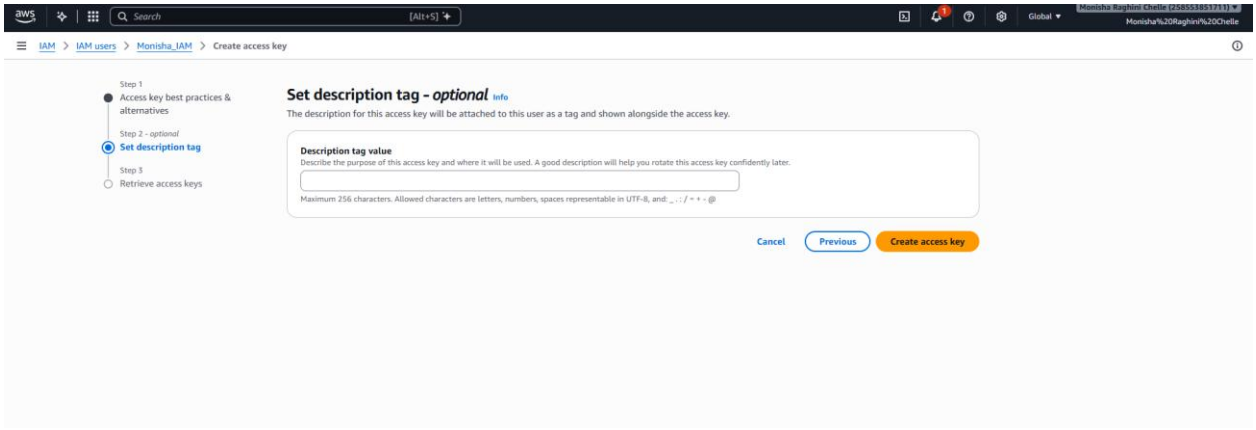
Description: *Opened the Create access key page and viewed the different use cases available for generating programmatic access credentials, such as CLI, local code and AWS applications.*



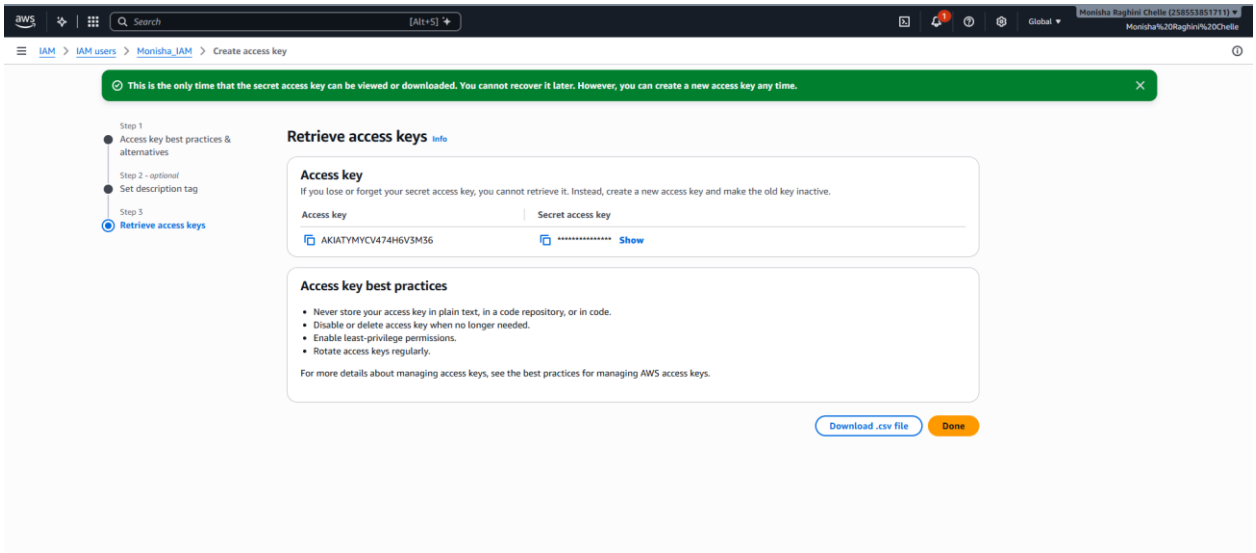
Description: *Selected the* ***Command Line Interface (CLI)*** *option as the access key use case. Reviewed the AWS security recommendations displayed on the page before proceeding with access key creation.*



Description: *Checked the confirmation box to acknowledge the AWS security recommendations and clicked* ***Next*** *to proceed with creating the access key using the CLI option.*

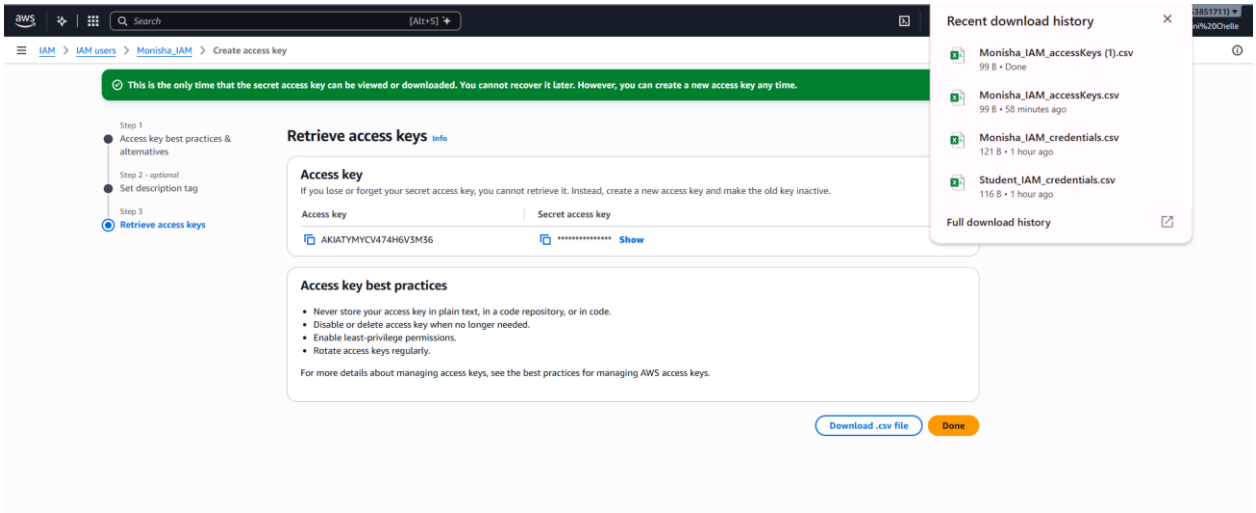


***Description**: Reached the Set description tag page, where an optional description can be entered to identify the purpose of the access key. Clicked **Create access key** to generate the credentials*

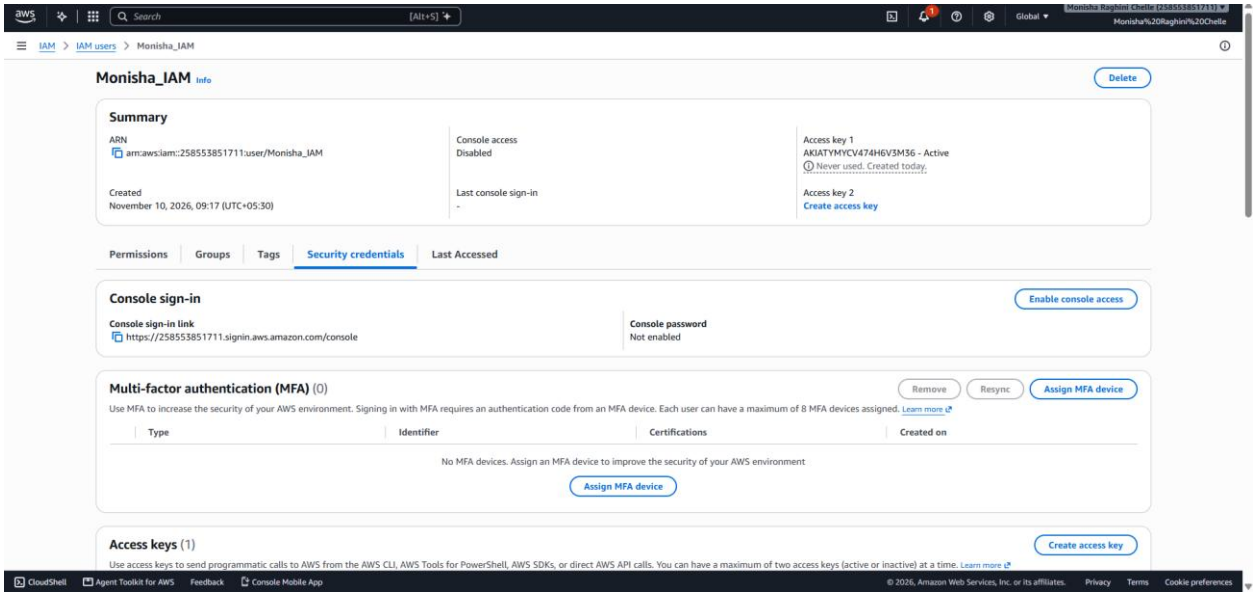


***Description:** The access key was successfully generated, and the Access Key ID and masked Secret Access Key are displayed on the page. The **Download .csv file** option is available to save the credentials securely.*

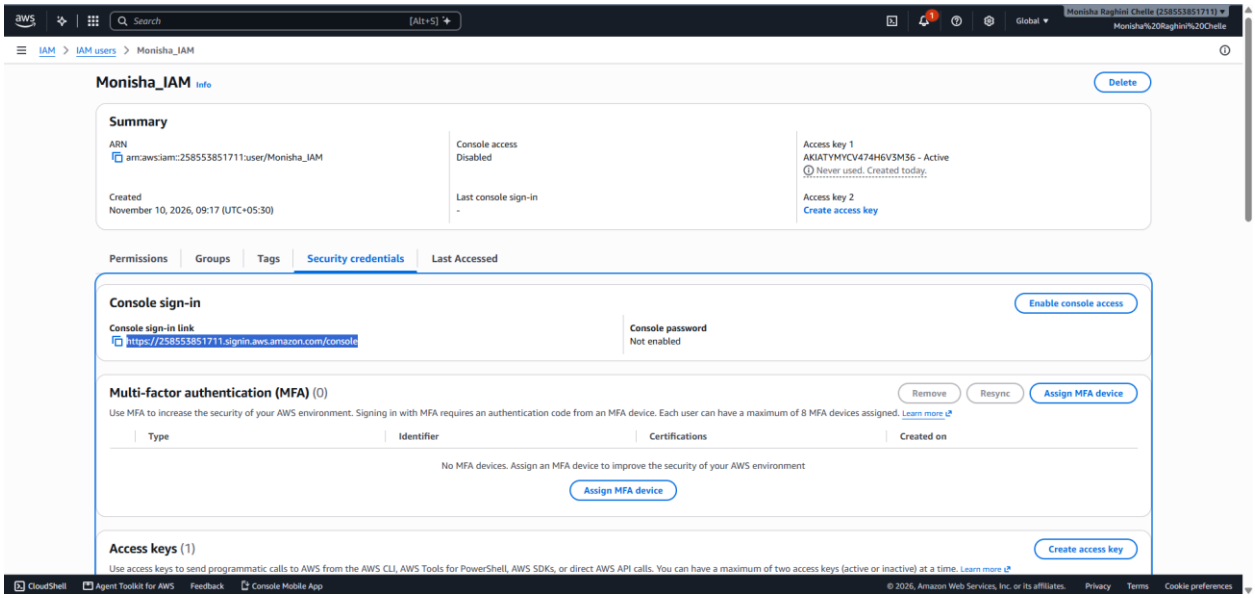
***Note:** Download the CSV file and keep the Secret Access Key safe, as AWS will not show it again after leaving this page.*



Description: *Downloaded the generated IAM access key credentials in CSV format. The browser's download history shows that the* ***Monisha_IAM_accessKeys.csv*** *file was downloaded successfully and saved.*

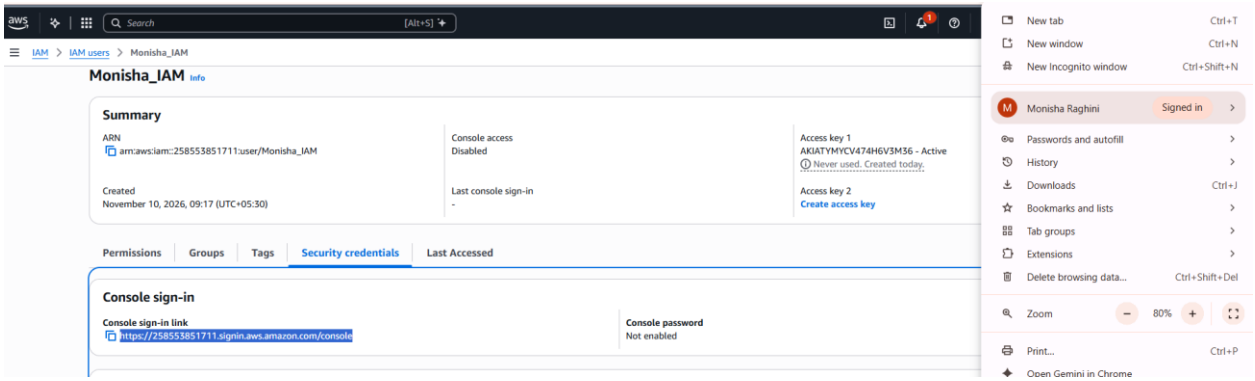


Description: *Returned to the IAM user's Security credentials page and verified that the generated access key is listed with an Active status. The console password remains disabled.*

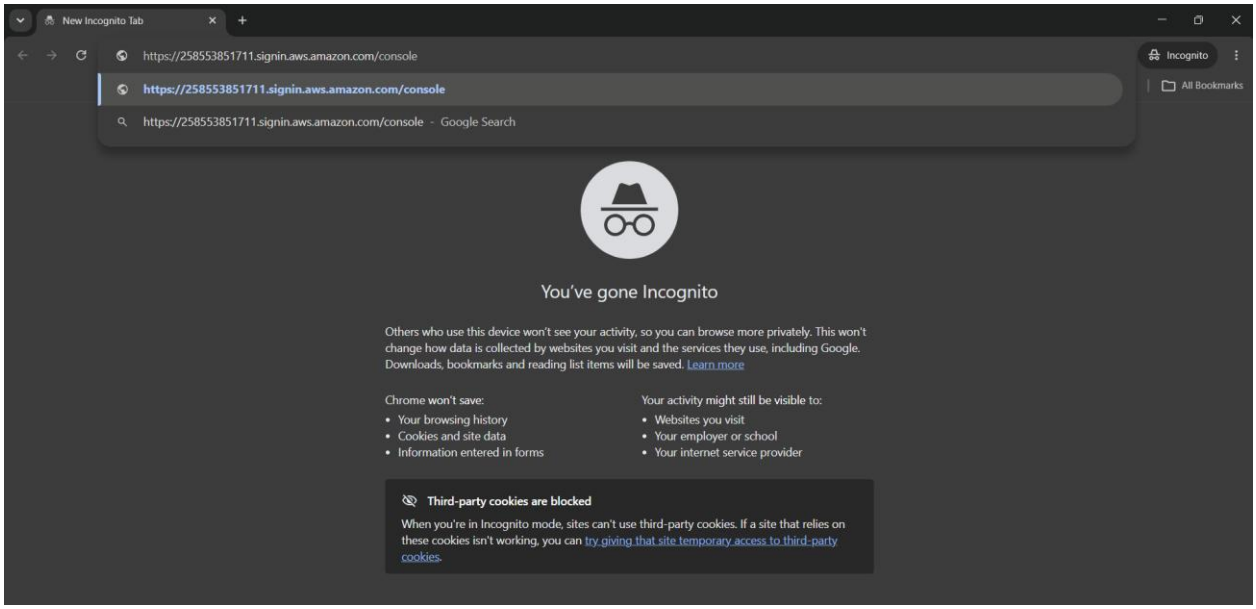


Description: *Selected the console sign-in URL on the Security credentials page and verified that the console password is not enabled. The page also shows that no MFA device has been assigned to the IAM user.*

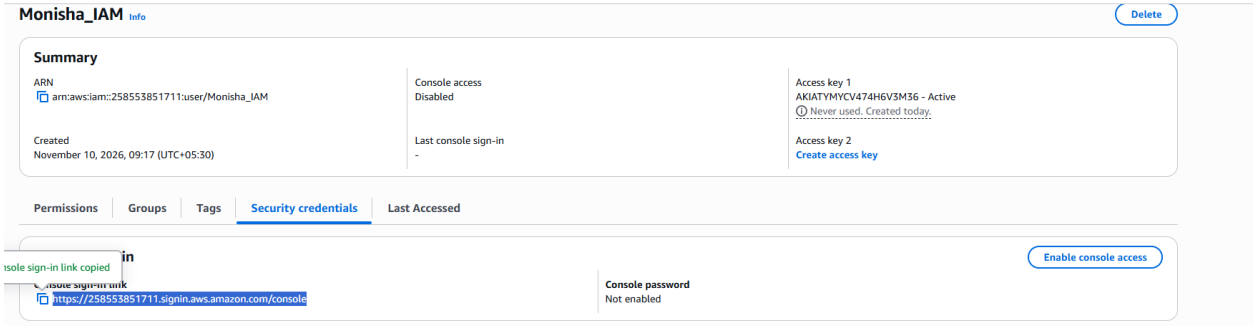
https://258553851711.signin.aws.amazon.com/console IAM sign-in URL:



Description: *Opened the Google Chrome menu and selected the* ***New Incognito window*** *option to access the AWS Management Console in a private browsing session.*



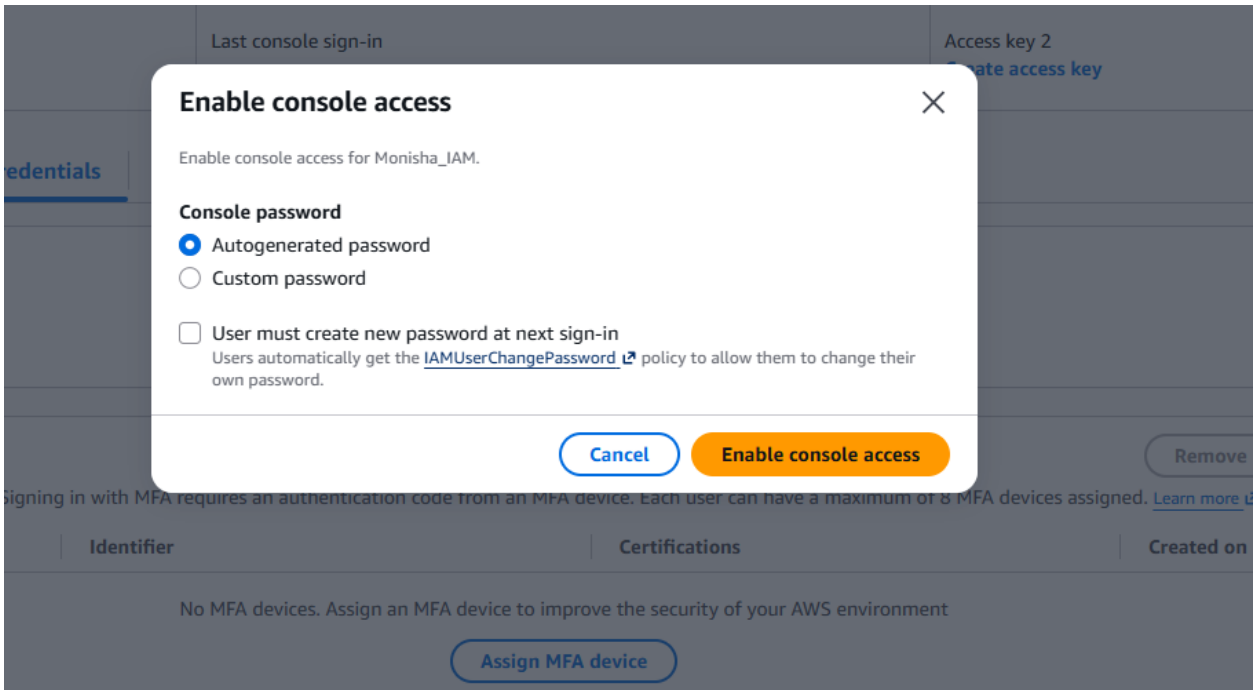
Description: *Opened a new Incognito window in Google Chrome and entered the AWS IAM user console sign-in URL in the address bar to access the IAM login page.*



Description*: Copied the IAM user's console sign-in URL from the Security credentials page. The page also shows that console access is disabled and the console password is not enabled.*

Note: This screenshot shows the URL being copied, not console access being enabled.

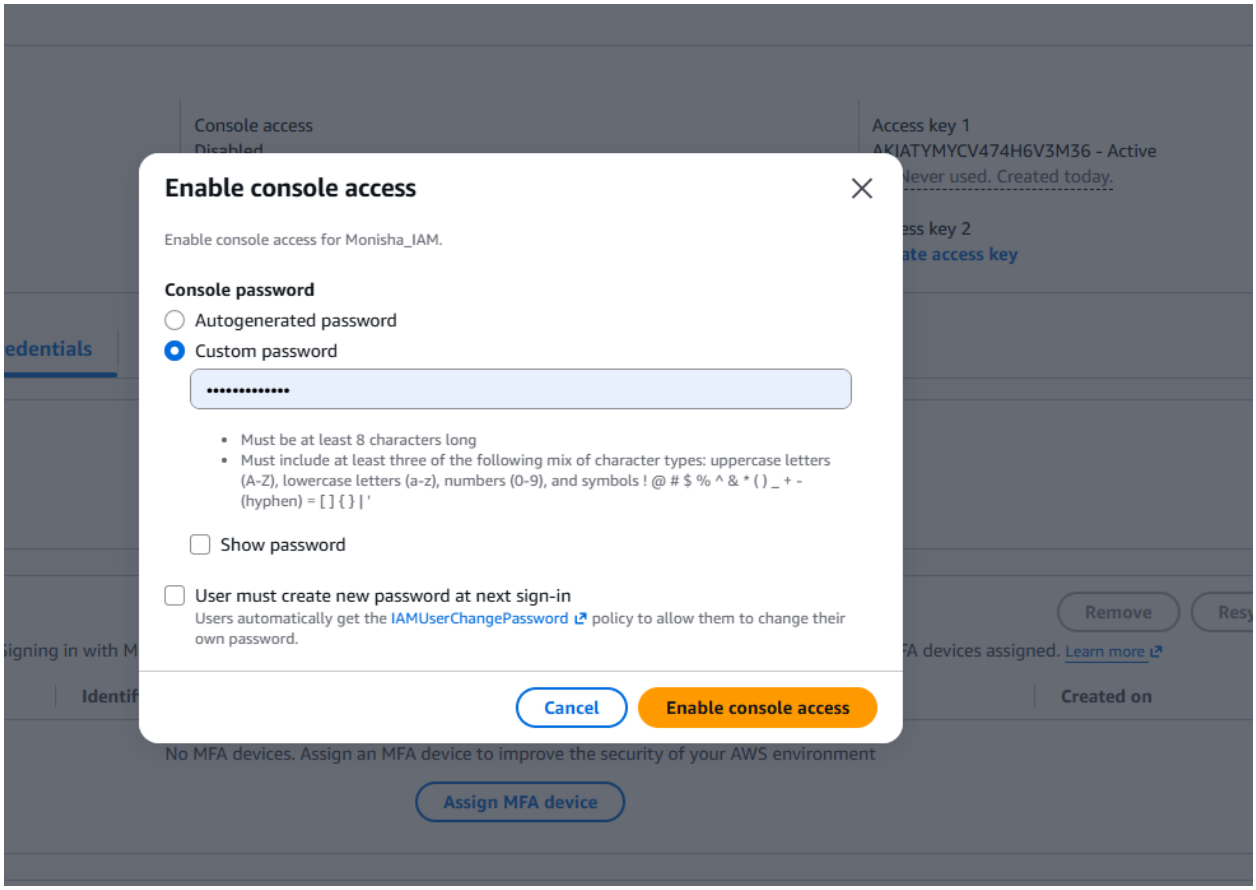
Click **Enable console access** to enable sign-in to the AWS Management Console.



Description: The IAM console access setup screen is displayed for the selected user.

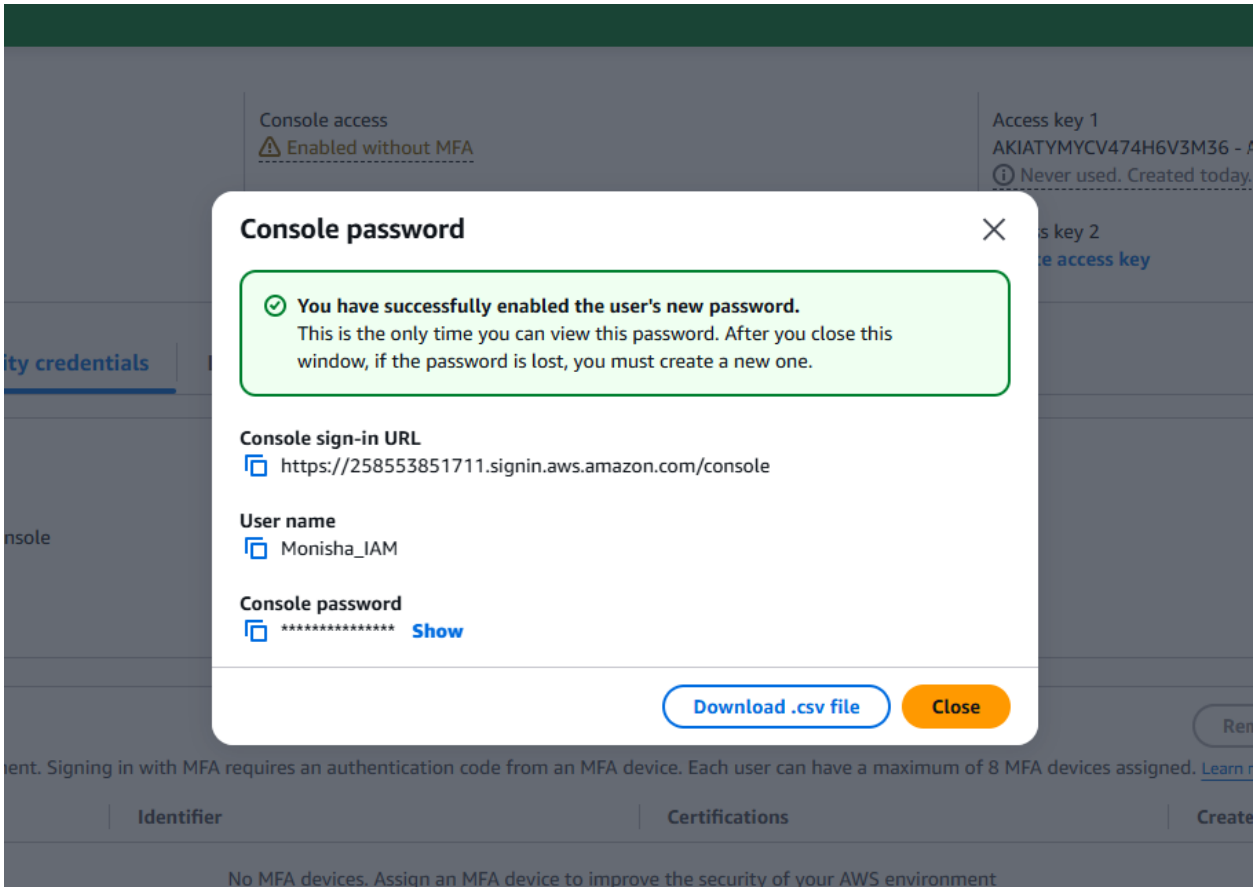
Description: *Selected the Custom password option under Console password to create a password manually instead of using the autogenerated password.*

Important: *This step enables AWS Management Console access.*



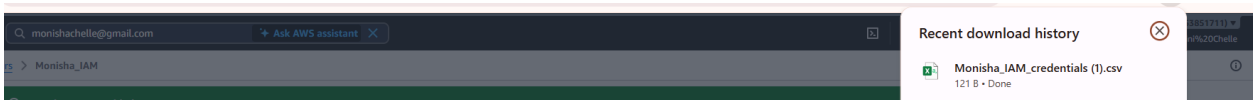
***Description**: Selected the Custom password option and entered a password of my choice in the password field. AWS displays the password requirements, including the minimum length and character combinations.*

***Important:** This screenshot is part of enabling console access.*



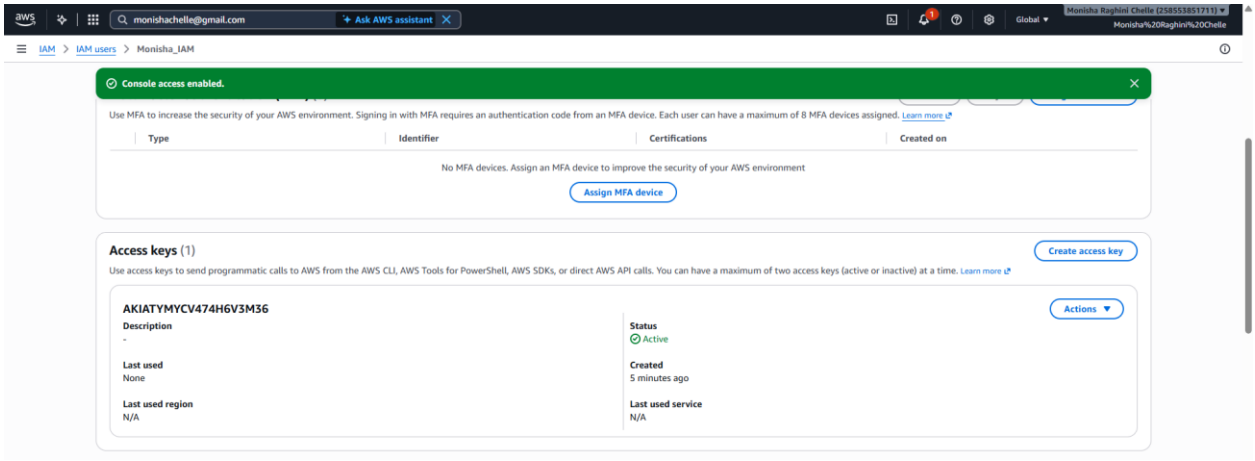
***Description**: Successfully enabled the IAM user's console password using the custom password option. AWS displayed a confirmation message along with the console sign-in URL, username and password download option.*

***Important:** This screenshot confirms that console access has been enabled without MFA**.*



***Description**: Downloaded the generated IAM console sign-in credentials in CSV format. The browser's download history shows that the **Monisha_IAM_credentials (1).csv file** was downloaded successfully.*

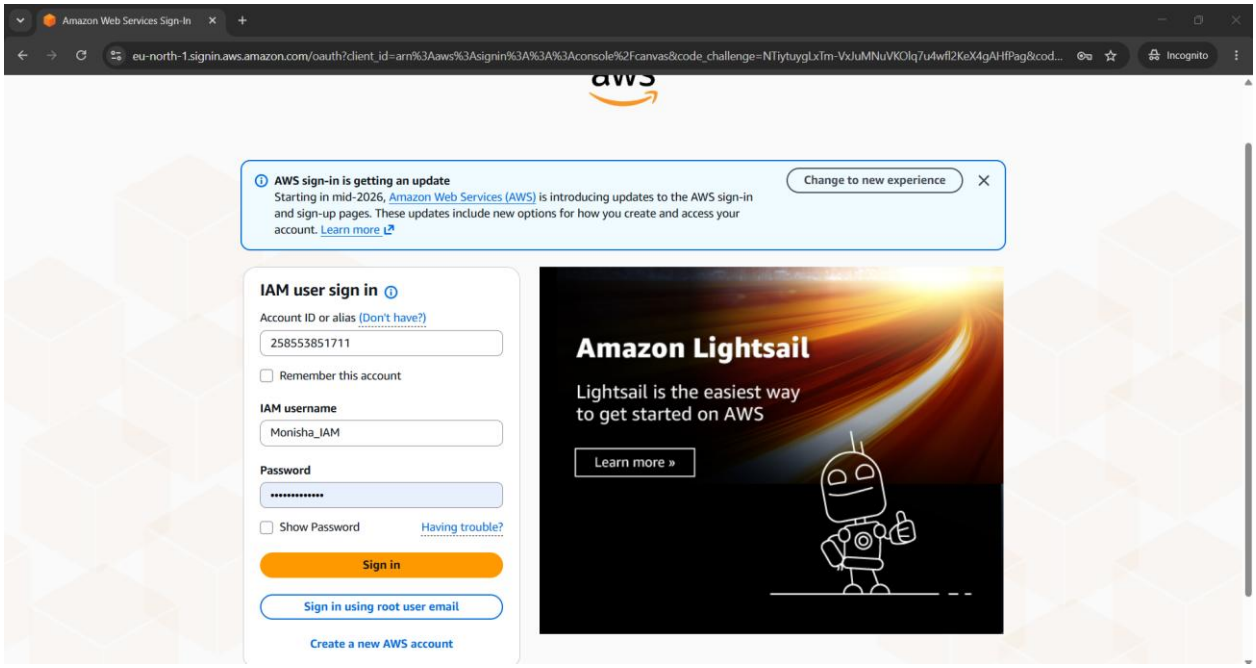
***Note:** This screenshot is related to console login credentials.*



Description: *Viewed the Access keys section under the IAM user's Security credentials tab and verified that the generated access key is listed with an Active status. The page also displays the key creation time and last-used details.*

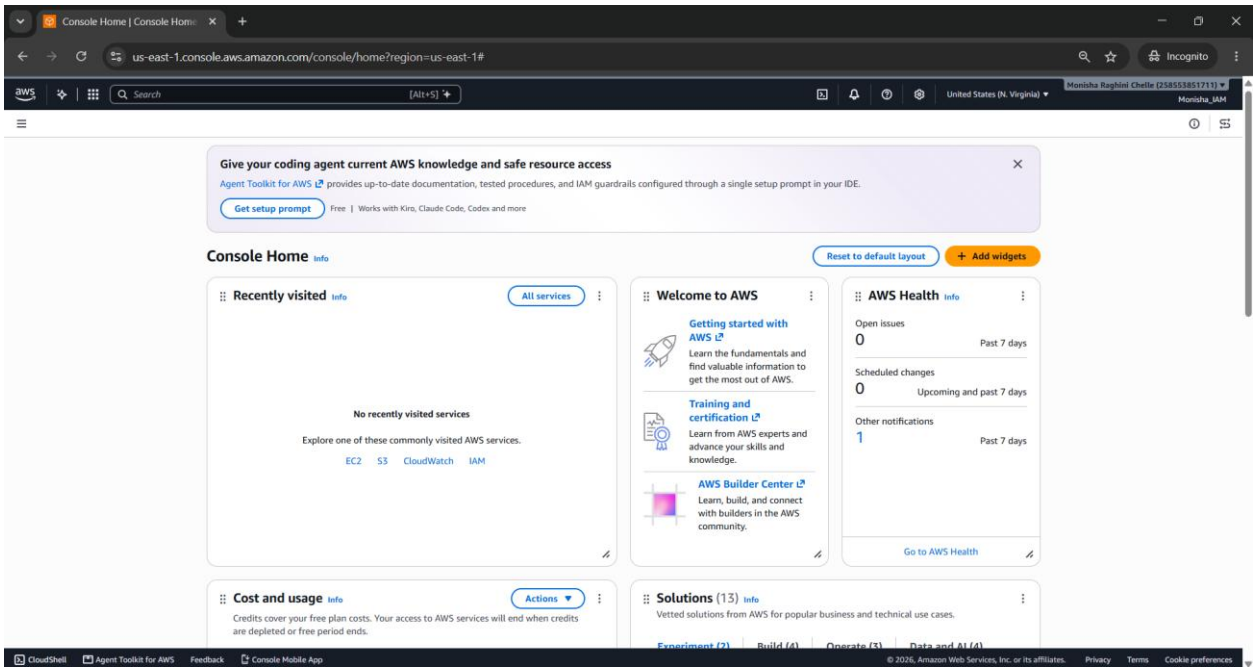
Important: *The green notification indicates that console access has been enabled.*

Incoginato

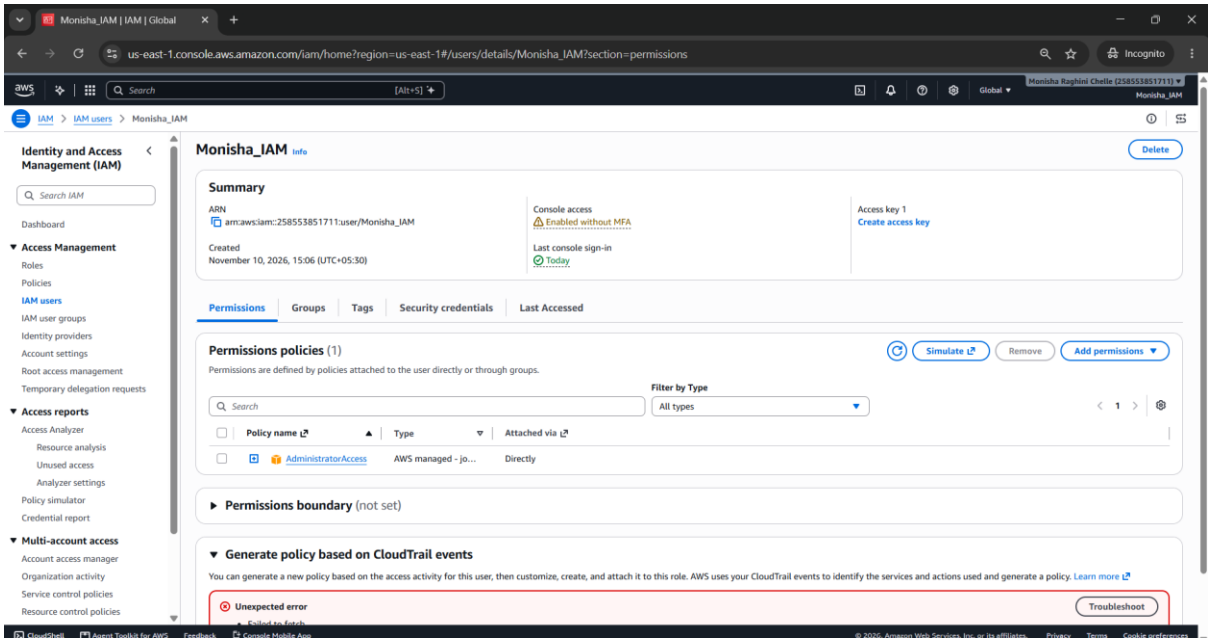


Description: *Opened the AWS IAM user sign-in page in a* ***new Incognito window****. Entered the AWS Account ID and IAM username* ***Monisha_IAM,*** *along with the configured console password, to sign in to the IAM user account.*

Important: *This screenshot shows console login.*

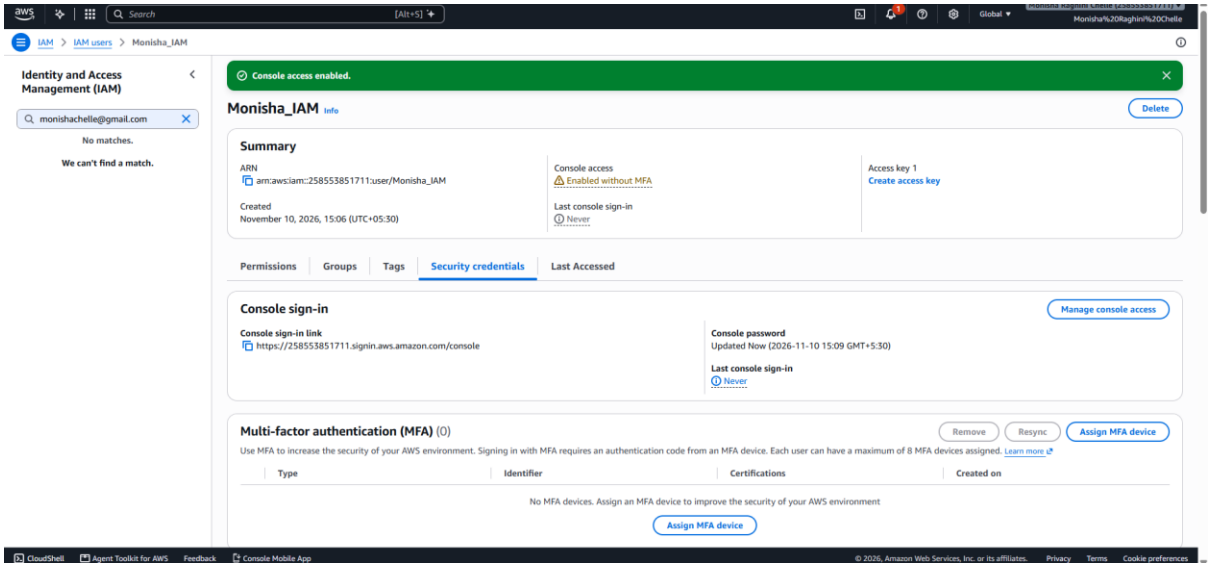


Description: *Successfully signed in to the AWS Management Console using the" **Monisha_IAM**" user account in Incognito mode. The AWS Console Home page is displayed after login.*

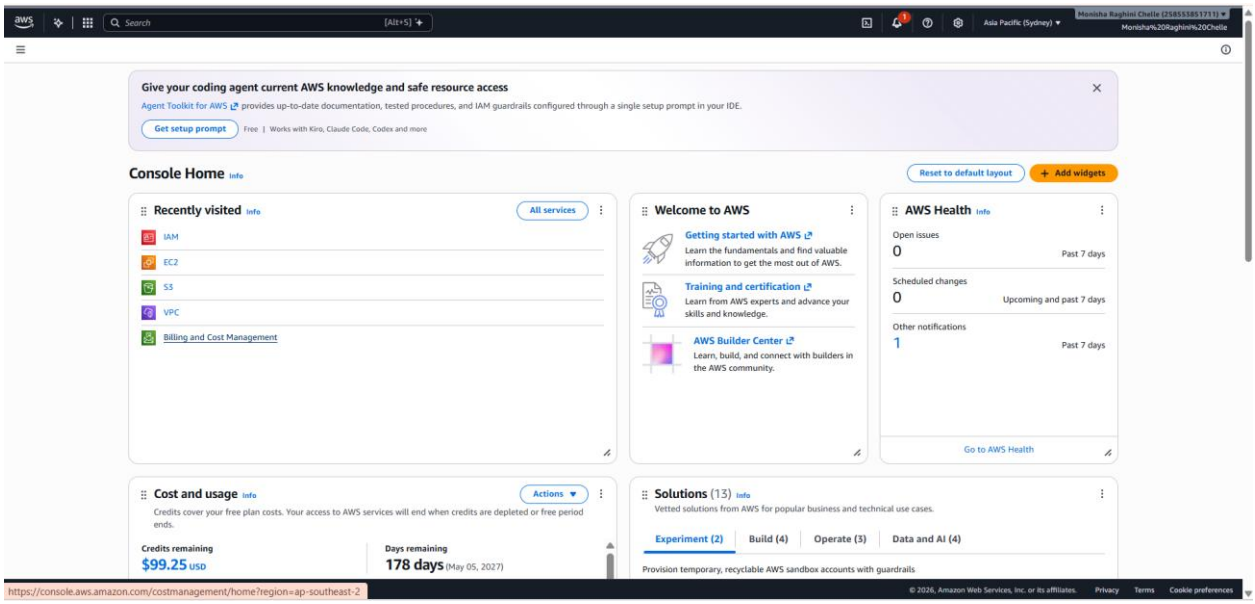


Description*: Opened the AWS IAM sign-in URL in Chrome Incognito and successfully logged in using the IAM user's credentials. The IAM dashboard confirms that console access is enabled and the AdministratorAccess policy is attached to the user*

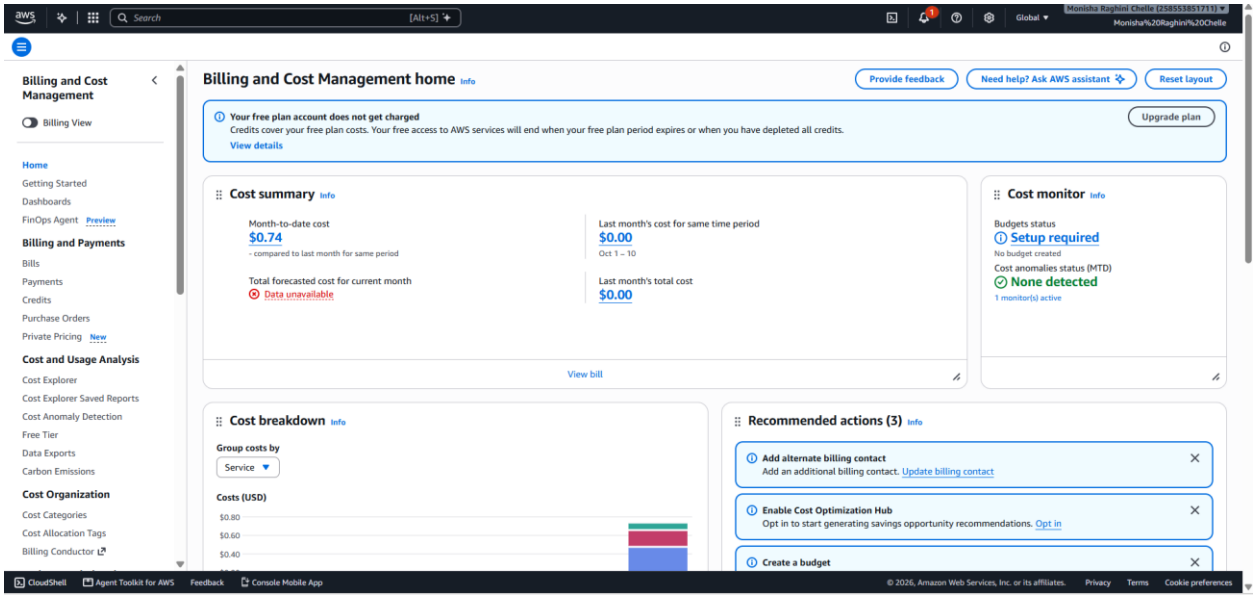
→IN GOOGLE CHROME



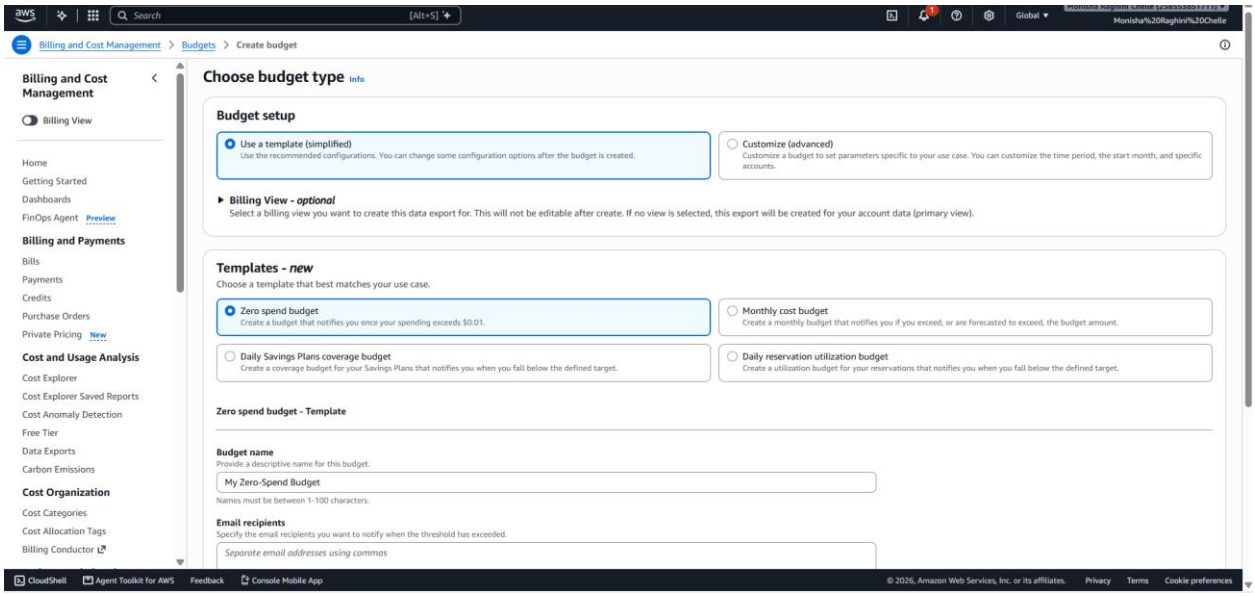
Description: *Returned to the IAM user's Security credentials page in Google Chrome and verified the console sign-in details, including the sign-in URL and console password status.*



Description: *Opened the AWS Management Console Home page in Google Chrome and clicked on Billing and Cost Management to access the account's billing details and cost information.*

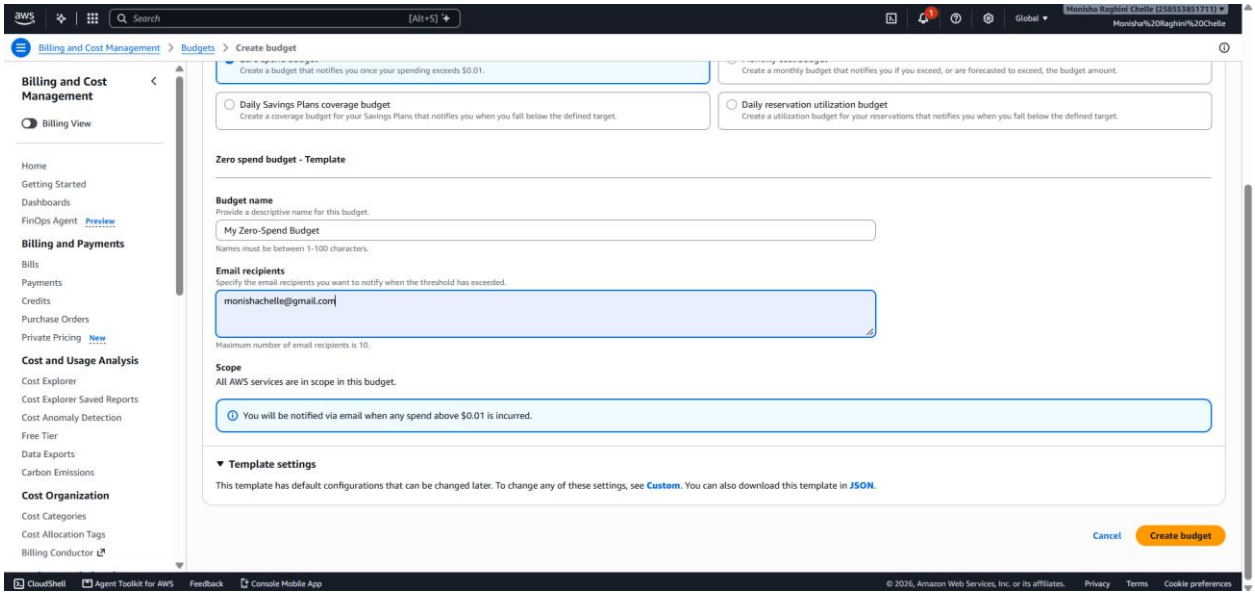


Description: *Clicked on the* ***Setup required*** *option under the Cost monitor section on the Billing and Cost Management dashboard to begin configuring AWS budget monitoring and alerts.*

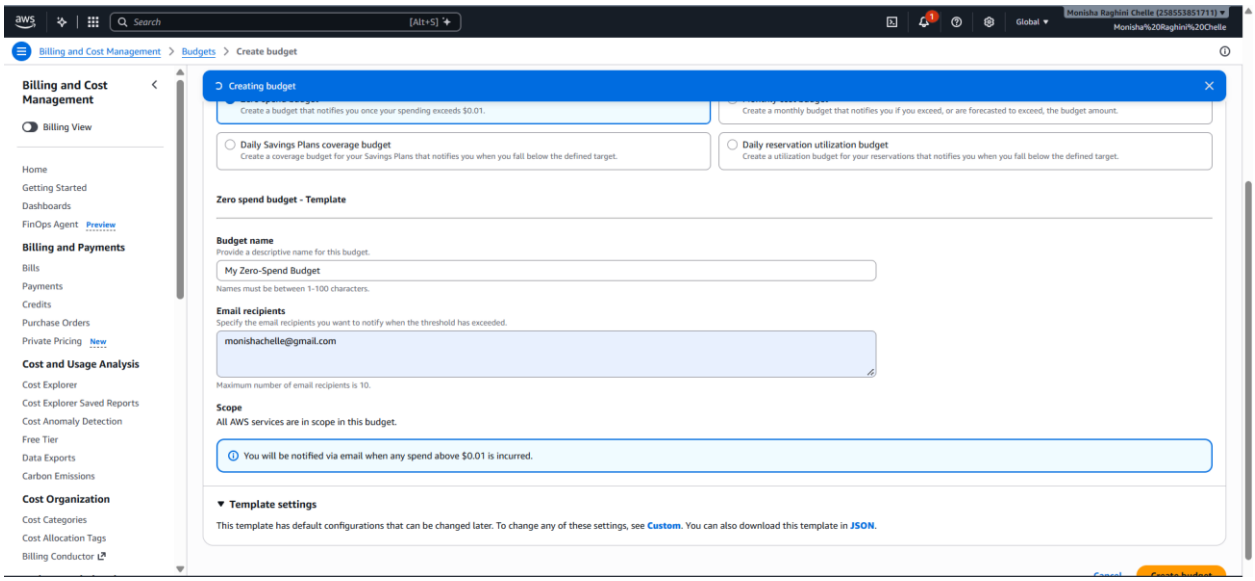


Description: *The Choose budget type page is displayed. I selected the* ***Use a template (simplified)*** *option and the* ***Zero spend budget*** *template. I entered a budget name and proceeded with the budget setup to receive an alert when AWS spending exceeds the specified limit.*

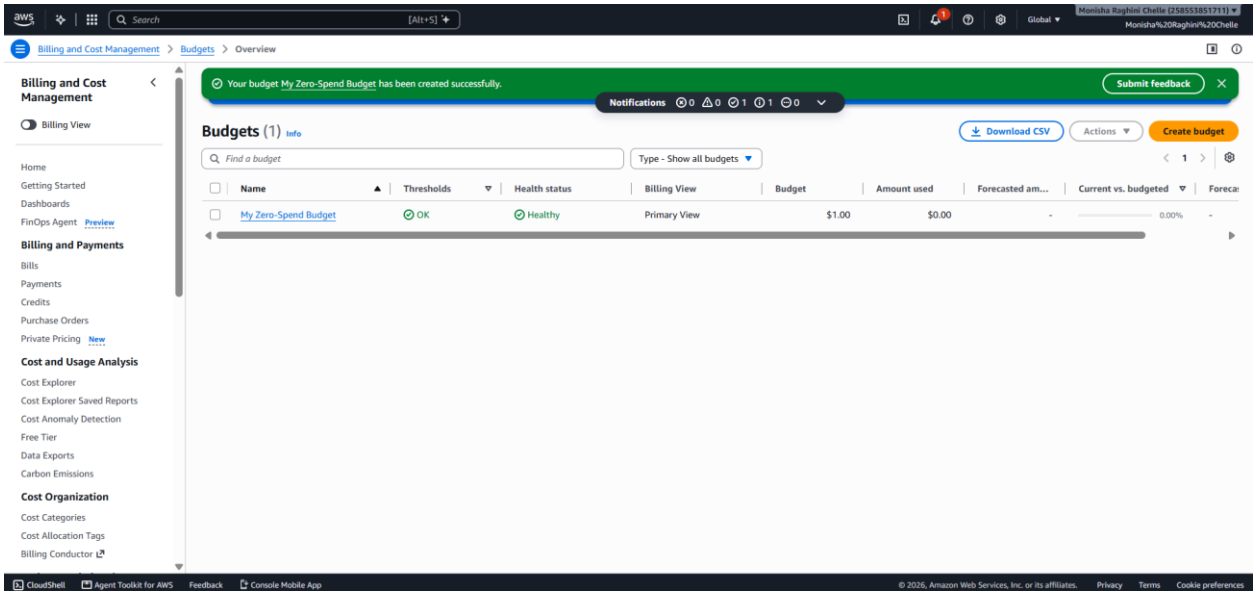
→Step: Enter Budget Details



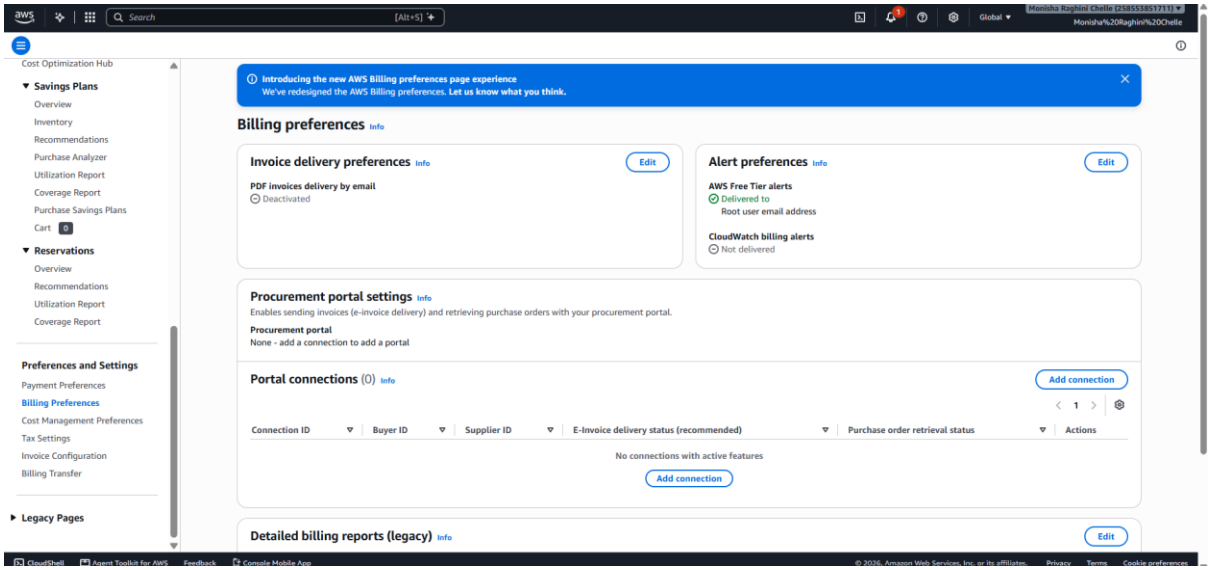
Description: *I selected the Zero spend budget template and entered the budget name. Then, I provided my email address (monishachelle@gmail.com) to receive notifications. This budget will send an email alert whenever my AWS spending exceeds $0.01.*



Description: *After entering the budget name and email address, I clicked the Create budget button. AWS started processing the budget creation request, as shown by the notification at the top of the screen.*



Description: *After clicking the Create budget button, AWS successfully created the Zero Spend Budget. The budget appears in the Budgets list with the name My Zero-Spend Budget. The budget amount is $1.00, and the amount used is currently $0.00. The health status is displayed as Healthy.*

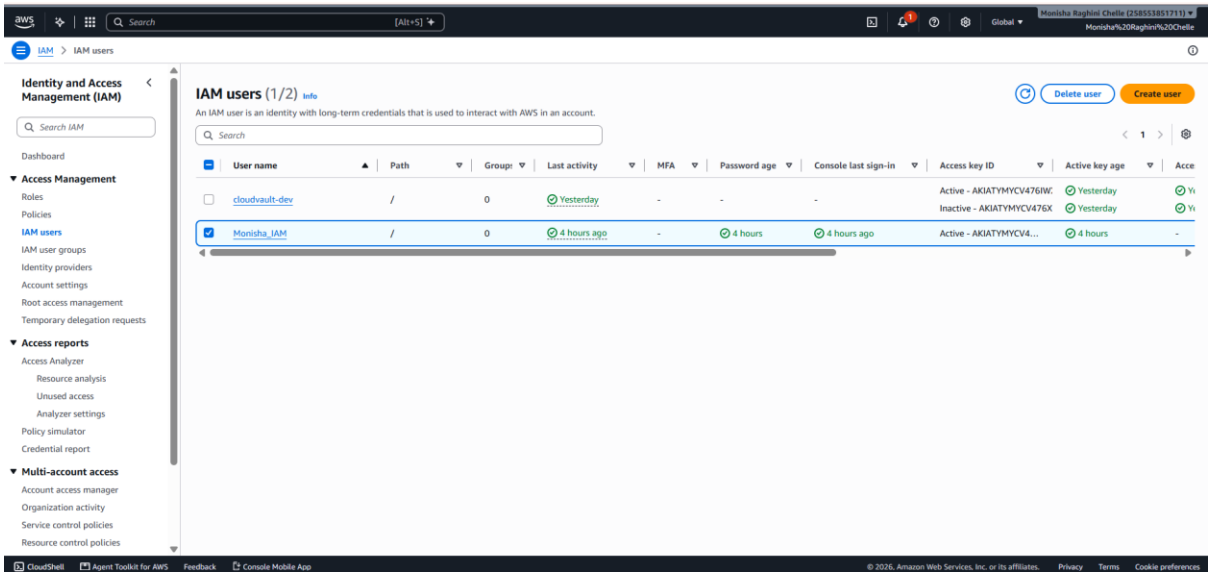


Description: Enabling AWS Free Tier Usage Alerts

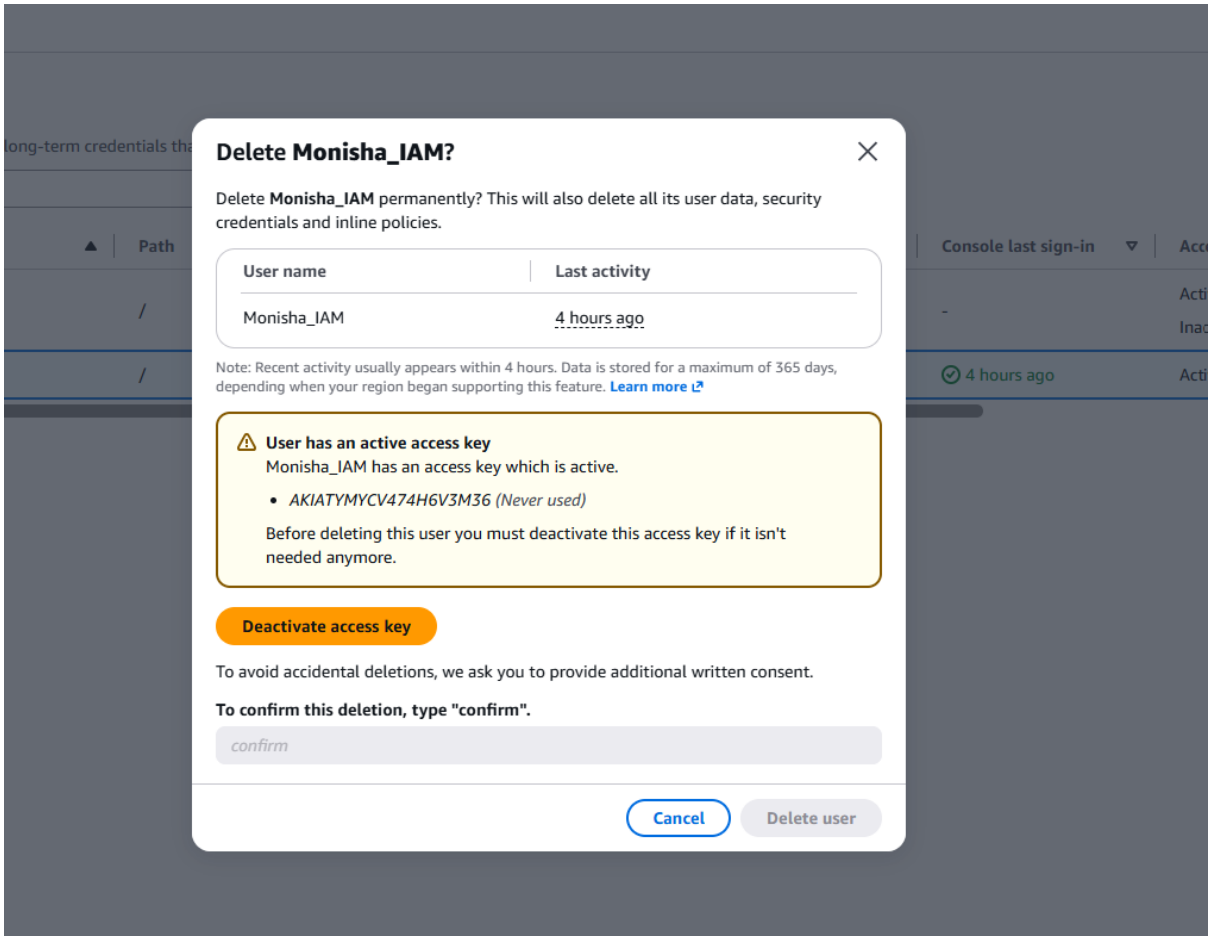
Navigated to the Billing Preferences section in AWS Billing and Cost Management. Under Alert Preferences, verified that AWS Free Tier alerts were successfully enabled and delivered to the root user email address. These alerts help monitor AWS Free Tier usage and notify the account owner when usage approaches or exceeds the applicable Free Tier limits.

Result: *AWS Free Tier usage alerts were successfully enabled and verified.*

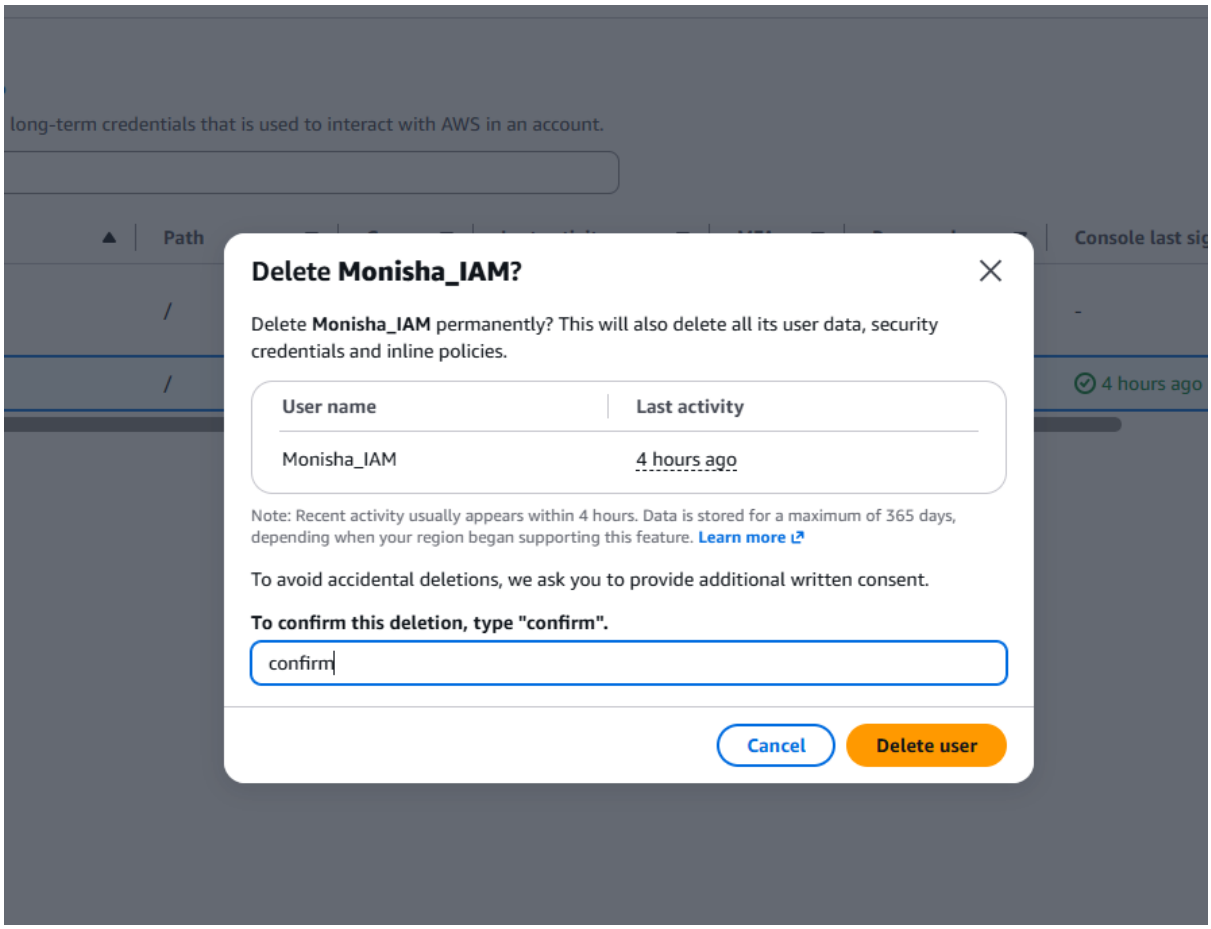
DELETE



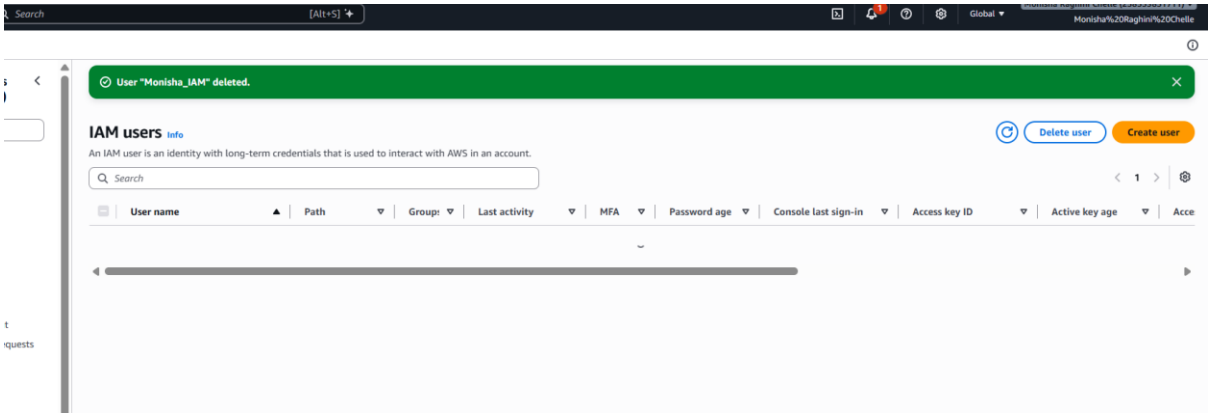
Description: *After completing the IAM user configuration, I selected the Monisha_IAM user from the IAM Users list. Before deleting the user, I deactivated its active access key. The access key status is now displayed as Inactive. This step ensures that the access key can no longer be used to access AWS services.*



Description: *When I clicked Delete user, AWS displayed a confirmation window for deleting the IAM user Monisha_IAM. It also showed a warning that the user still had an active access key. AWS requires the active access key to be deactivated before the user can be permanently deleted.*



Description: *After deactivating the access key, I entered the word confirm in the confirmation box to approve the permanent deletion of the IAM user Monisha_IAM. The **Delete user** button became enabled, allowing me to proceed with the deletion.* Clicked the orange **Delete user** button to complete the deletion.



Description: *After clicking the Delete user button, AWS successfully deleted the IAM user Monisha_IAM. A green confirmation message appeared at the top of the screen stating that the user was deleted. The IAM Users list no longer displays the deleted user.*

Steps:

Part A: IAM User Creation and Programmatic Access

1. Log in to the AWS Management Console using the root account.
2. Search for IAM in the AWS search bar and open the IAM service.
3. Select Users from the left navigation panel.
4. Click Create user to create a new IAM user.
5. Enter the username Monisha_IAM and initially keep AWS Management Console access disabled.
6. Select the required permissions. In this procedure, the AdministratorAccess policy was attached directly to the user.
7. Review the user details and click Create user.
8. Select the newly created IAM user from the Users list.
9. Open the Security credentials tab.
10. Navigate to the Access keys section and click Create access key.
11. Select Command Line Interface (CLI) as the intended use case.
12. Review and acknowledge the security recommendations.
13. Enter an optional description tag and proceed to create the access key.
14. Retrieve the generated Access Key ID and Secret Access Key.
15. Download the access credentials in CSV format and store them securely.
16. Verify that the access key is active.

Part B: AWS Billing Alert Configuration

17. Open Billing and Cost Management from the AWS Management Console.
18. Navigate to the Billing and Cost Management dashboard.
19. Locate the Cost Monitor section.
20. Click Setup required to configure a budget.

21. Select Use a template (simplified).
22. Select the Zero spend budget template.
23. Enter the budget name as My Zero-Spend Budget.
24. Enter the email address to receive budget notifications.
25. Review the notification threshold, which is triggered when spending exceeds $0.01.
26. Click Create budget to save the configuration.
27. Verify that the budget appears in the Budgets list.
28. Confirm that the budget status is Healthy.
29. Navigate to Billing Preferences and verify that AWS Free Tier alerts are enabled and delivered to the root user email address.

Output:

1. Successfully created an IAM user named **Monisha_IAM.**
2. Assigned the **AdministratorAccess** policy to the IAM user.
3. Selected CLI as the intended use case for programmatic access.
4. Generated the Access Key ID and Secret Access Key.
5. Downloaded the generated access credentials in CSV format and stored them securely.
6. Verified that the generated access key was active.
7. Opened AWS Billing and Cost Management.
8. Successfully created a Zero Spend Budget named My Zero-Spend Budget.

9. Configured the email address to receive budget notifications when spending exceeds $0.01.
10. Verified that the newly created budget appeared in the AWS Budgets dashboard with a Healthy status.
11. Successfully verified that AWS Free Tier usage alerts were enabled and delivered to the root user email address.

Result:

The IAM user named Monisha_IAM was successfully created with the required AdministratorAccess permissions. Programmatic access credentials (Access Key ID and Secret Access Key) were generated and downloaded in CSV format. The access key was verified as active. Additionally, a Zero Spend Budget named My Zero-Spend Budget was successfully created in AWS Billing and Cost Management, and email notifications were configured for spending exceeding $0.01. AWS Free Tier usage alerts were also verified as enabled and delivered to the root user email address. After completing the practical, the access key was deactivated and the IAM user was successfully deleted.